

2.6 Exercise Sheet 6

2.6.A Numerical and Exploration

Exercise 2.6.1.

- (a) In 1.22.13, we showed that there are no four-term sequences in arithmetic progression comprised of integer squares (although there are infinitely many such three-term sequences which are even coprime, by 2.3.4). What can you say about sequences of integer cubes in arithmetic progression? What about integer fourth powers?
- (b) Here's an open-ended follow-up: what can you say about squares in arithmetic progression over other fields? Consider the case of finite fields $K = \mathbb{F}_q$ for $q = 2, 3, 4, 5, 7, 9, 11$, and the case of the number fields $K = \mathbb{Q}[\sqrt{409}]$ and $K = \mathbb{Q}[\beta]$ where $\beta^3 - \beta^2 + 1 = 0$. Suggested further reading: this article [21] by Xarles for an original investigation, and this article [22] by González-Jiménez (and the references cited therein) for the current state of knowledge on this topic (as of this writing in 2026).

Exercise 2.6.2. Let $m, n \in \mathbb{Z}_{\geq 0}$, and let M be an $m \times n$ integer matrix. For an $r \in \mathbb{Z}_{\geq 0}$ with $r \leq \min\{m, n\}$, by an $r \times r$ -minor of M we mean the determinant of a square submatrix of M of size $r \times r$. For each such r , let $B_r(M) \subset \mathbb{Z}$ be the subgroup generated by all the $r \times r$ -minors of M .

- (a) For a fixed matrix M , what is the containment relationship between the various $B_r(M)$ s as r varies? Make and prove conjectures.
- (b) Suppose we put M into Smith Normal Form to obtain $(t; b_1, \dots, b_t)$ as in 1.21.6. For each r , what is the relationship between the integer b_r and the subgroup $B_r(M) \subset \mathbb{Z}$? Work out lots of examples, and make and prove conjectures. Show in particular that these invariants $(t; b_1, \dots, b_t)$ are uniquely determined by M .

Exercise 2.6.3. For a prime p , let $E_p : y^2 = x^3 + px$ be the elliptic curve over $\mathbb{Q}$ as in 1.22.14

- (a) Show that for each prime p , we have that $\text{rank } E_p \in \{0, 1, 2\}$. Do all three possibilities occur?
- (b) Compute the rank of E_2 (by hand, as done in 1.22.14).
- (c) (*) Finish the computation of the rank of E_{17} started in class by determining whether the homogeneous spaces $w^2 = -2u^4 + 34v^4$ and $w^2 = 17u^4 - 4v^4$ have any nontrivial rational points.
- (d) Using Sage, compute the ranks of E_p for all primes $p \leq 10000$. (Hint: You may have to use `E.rank(only_use_mwrank = False)`.)
- (e) What patterns do you notice in this table? For instance, what can you say about the relationship between $\text{rank } E_p$ and the congruence class of $p \pmod{16}$? (Hint: Study the groups of congruence classes $\{7, 11\}$ and $\{3, 5, 13, 15\}$ and $\{1, 9\}$ separately.) Make conjectures.
- (f) (*) Prove some of your conjectures from part (e). For instance, formulate a conjecture in the case $p \equiv 7, 11 \pmod{16}$ and prove it. Similarly, show that if $p \equiv 3, 5, 13, 15 \pmod{16}$, then $\text{rank } E_p \leq 1$.

Exercise 2.6.4. We saw in 1.22.14 that the elliptic curve $E_5 : y^2 = x^3 + 5x$ over $\mathbb{Q}$ has rank 1. In this exercise, we will figure out how to determine explicit generators of the Mordell-Weil group $E_5(\mathbb{Q})$.

- (a) Use Nagell-Lutz (1.13.7) to determine $\text{Tors } E_5(\mathbb{Q})$.

In the notation of 1.22.14 we had found that the point $(u, v, w) = (1, 1, 1)$ lies on the homogeneous space $w^2 = 5u^4 - 4v^4$ corresponding to the $b_1 = 5$ on the two-isogenous curve $E'_5 : y^2 = x^3 - 20x$. Using 1.21.2 this corresponds to the rational point $(5, 5) \in E'_5(\mathbb{Q})$. Applying ϕ' to this gives us a point $(1/4, 9/8) \in E_5(\mathbb{Q})$, which after translation by $T = (0, 0)$ (see 1.20.10) gives us the points $\pm P = (20, \pm 90) \in E_5(\mathbb{Q})$. For the sake of concreteness, let $P := (20, 90)$. Note that, according to your answer from (a), we have that $P \in E_5(\mathbb{Q}) \setminus \text{Tors } E_5(\mathbb{Q})$ (this also follows from $P + T = (1/4, -9/8)$ thanks to 1.13.7, so this means that P is a multiple of a generator of the free part of $E_5(\mathbb{Q})$). We will prove that P is a generator of the free part of $E_5(\mathbb{Q})$, which we combined with (a) will give us a complete description of $E_5(\mathbb{Q})$.

- (b) Make the bound in 1.17.9(a) explicit. In other words, find an explicit constant $C \in \mathbb{R}_{\geq 0}$ such that for all $Q \in E_5(\mathbb{Q})$, we have that $|\hat{h}(Q) - h(Q)| \leq C$. (Hint: Trace through the proof to see how to produce such a C . You may not have to work through the entirety of 2.5.5 you can just try to use the bound from 1.16.6(c) along with a very naive upper bound. It should suffice to take C to be a very small positive integer.)
- (c) Suppose that P is not a generator, so that there is a $Q \in E_5(\mathbb{Q})$ and an integer $m \in \mathbb{Z}_{\geq 2}$ such that

$P = mQ$. Using the fact that $\hat{h}$ grows quadratically and using your explicit bound from (b), show that the *naïve* heights of P and Q are related by

$$h(Q) \leq \frac{1}{4}h(P) + \frac{5C}{4}.$$

- (d) Write a simple program in Sage to manually search for all points Q on $E_5(\mathbb{Q})$ of naive height at most the bound found in (c). Use this to prove that P is indeed the generator of the Mordell-Weil group.

A similar technique (with various efficiency enhancements) can be used in general to verify whether a given point is a Mordell-Weil generator; combined with the result of 2.5.3(d), this technique is often successful in practice at finding a complete basis of the Mordell-Weil lattice after having performed two-descent.

Exercise 2.6.5. For $N \in \mathbb{Z}_{\geq 0}$, let $C \subset \mathbb{P}_{\mathbb{Q}}^2$ be the cubic curve of the “fruit meme” obtained by clearing denominators in the equation

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} = 4$$

as in lecture.

- (a) Show that $[1 : -1 : 0] \in C(\mathbb{Q})$ is a rational inflection point. Use 1.9.6 to put C into Weierstrass form $E : y^2 = x^3 + 109x^2 + 224x$. Use Nagell-Lutz 1.13.7 to compute $\text{Tors } E(\mathbb{Q})$.
- (b) If you’re feeling energetic, use two-descent as in 1.22.B to compute $\text{rank } E$, and then use the argument of 2.6.4 to find explicit generators of $E(\mathbb{Q})$. If you’re feeling less energetic (but still enthusiastic about this question), ask Sage to do this for you. In any case, get that $P = (-4, 28)$ is a generator of $E(\mathbb{Q})$.
- (c) Show that one of the points $9P + Q$ for $Q \in \text{Tors } E(\mathbb{Q})$ gives rise to a solution a, b, c of the above equation with $a, b, c \in \mathbb{Z}_{\geq 1}$. (Hint: These computations are very difficult (impossible?) to do by hand, but Sage handles them quite well.)
- (d) Here’s an open-ended follow-up: can you repeat the same procedure with 4 replaced by an arbitrary $N \in \mathbb{Z}_{\geq 1}$? See how far you can get. Suggested further reading: 23 for a detailed discussion of this problem, and 24 for many such similar “recreational” examples.

Exercise 2.6.6 (RSA Cryptosystem). (Adapted from 2 Exercise 4.25) This exercise explains the basic idea behind the RSA cryptosystem.

Suppose that Alex would like to send Billie messages over a channel that can be invaded by Charlie, as in class. To do this, Alex and Billie agree beforehand on two large primes p, q ; these primes are kept secret. Consider the integers $N = pq$ and e , where $1 < e < (p-1)(q-1)$ and $\gcd(e, (p-1)(q-1)) = 1$; the integers N and e are released to the public.

- (a) Alex encrypts the desired message, expressed via a predetermined padding scheme to be an element $m \in \mathbb{Z}/N\mathbb{Z}$, by computing the value of the ciphertext $c \in \mathbb{Z}/N\mathbb{Z}$ given by $c \equiv m^e \pmod{N}$. Then Alex sends c to Billie along the possibly insecure channel. Show that if Billie knows the values of p and q , then she can efficiently find an integer f such that for all $m \in \mathbb{Z}/N\mathbb{Z}$ we have $m^{ef} \equiv m \pmod{N}$, and use this to recover the message m from c . The idea is that it will be very hard for Charlie to find the right value of f which “undoes the encryption” unless Charlie knows p and q ; in other words, the security of the RSA cryptosystem relies on the difficulty of the factoring problem for very large integers N .
- (b) Show that the knowledge of p and q is equivalent to the knowledge of $N = pq$ and $(p-1)(q-1)$ in the sense that each pair of integers can be recovered from the other. In particular, if Charlie knows $(p-1)(q-1)$, then Charlie can easily decrypt the message.

Exercise 2.6.7. (Adapted from 2 Exercise 4.21) Suppose you are given a big integer $N \in \mathbb{Z}_{\geq 1}$ that you would like to factor using Lenstra’s algorithm. As explained in class, the basic idea is to consider the point $P = (1, 1)$ on the “elliptic curve” over $\mathbb{Z}/n\mathbb{Z}$ given by $E_b : y^2 = x^3 + bx - b$ for various $b = 1, 2, \dots$. Using the extremely efficient double-and-add algorithm described in class, we can then compute multiples of points on E_b over $\mathbb{Z}/N\mathbb{Z}$, so suppose we try to compute say $(20!) \cdot P$ in E_b . If the computations go through, we pick the next b and repeat. If the computation breaks down for some step, then we have (hopefully!) found a factor of N .

Write simple Sage code to implement this in practice, and use this to factor the integers $N = 199843247$ and $N = 4686706420729$. (Of course, Sage can just factor these numbers very quickly. The point of this exercise is to see how Lenstra’s algorithm might be implemented in practice.)

2.6.B PODASIPs

Prove or disprove and salvage if possible the following statements.

Exercise 2.6.8. Let K be a field and $C \subset \mathbb{P}_K^2$ be an integral plane curve. Then C is *generically smooth*, i.e., for every field extension L/K , all but finitely many points of $C(L)$ are smooth points. (Hint: Consider $\mathbb{V}(X^2 + tY^2) \subset \mathbb{P}_{\mathbb{F}_2(t)}^2$. For a strong salvage, consider the case of perfect K or geometrically integral C .)

Exercise 2.6.9. Let K be a field and let $a_1, \dots, a_6 \in K$ be any elements (in particular, we don't require that $\Delta \neq 0$). Let $E \subset \mathbb{P}_K^2$ be the curve given by [1.9.3](#). Then E is geometrically integral. (Hint: Use [2.3.6](#) to reduce to showing that a polynomial $f(x, y) \in K[x, y]$ is irreducible. Note that $f(x, y)$ is quadratic in y . Find general criteria for when such a polynomial is irreducible, possibly by invoking Gauss's Lemma to pass between $K[x, y]$ and $K(x)[y]$.)

Exercise 2.6.10. Let K be a perfect field and $C \subset \mathbb{P}_K^2$ a geometrically integral cubic curve. We investigate what singularities of C can look like. Suppose C is singular. By [2.2.7](#) there is a unique singular point $P \in C(K)$. By a projective change of coordinates, we can take P to $[0 : 0 : 1]$.

- (a) (**) There is a projective change of coordinates which brings C into exactly one of the following two forms

$$C_a : Y^2Z = X^3 \quad \text{or} \quad C_{m,\alpha} : Y^2Z = X^3 + \alpha X^2Z$$

for some $\alpha \in K^\times$. (Hint: This result is incorrect as stated in characteristics 2 and 3. You may also need to consider the "Artin-Schreier cubic" $Y^2Z + XYZ + X^2Z = X^3$ in characteristic 2 and the "strange cusp" $Y^2Z = X^3 + X^2Y$ in characteristic 3.)

A curve which can be brought into the form C_a is said to be *cuspidal* and one which can be brought into the form $C_{m,\alpha}$ *nodal*; if we can further make $\alpha = 1$, then the curve is called *split nodal*, and if we cannot then it is called *nonsplit nodal*.

- (b) Whether or not the nodal curve $Y^2Z = X^3 + \alpha X^2Z$ is split depends only on the class of α in $K^\times / (K^\times)^2$. Every nodal curve over an algebraically closed field is split. The curve $Y^2Z = X^3 - X^2Z$ is a nonsplit nodal curve over $\mathbb{Q}$. (Hint: Again, something funny is going on when $\text{ch } K = 2$.)
- (c) Consider the cuspidal curve $C_a = \mathbb{V}(Y^2Z - X^3) \subset \mathbb{P}_K^2$. Look at the chart $Y \neq 0$ with coordinates $(t, s) = (X/Y, Z/Y)$ in which C_a looks like $s = t^3$. The map $t : C_a(K) \setminus \{P\} \rightarrow K$ is an isomorphism of abelian groups when the left hand side is given the group law discussed in the lecture.
- (d) Consider the split nodal curve $C_m = C_{m,1} = \mathbb{V}(Y^2Z - X^3 - X^2Z) \subset \mathbb{P}_K^2$ when $\text{ch } K \neq 2$. As in (b), we get the affine model $s = t^3 + t^2s$. The map $C_m(K) \setminus \{P\} \rightarrow K^\times$ given by

$$(t, s) \mapsto \frac{1-t}{1+t}$$

is a group isomorphism.

- (e) (**) Finally, consider the non-split nodal curve $C_{m,\alpha} = \mathbb{V}(Y^2Z - X^3 + \alpha X^2Z) \subset \mathbb{P}_K^2$ when $\text{ch } K \neq 2$ and $\alpha \neq [1] \in K^\times / (K^\times)^2$. Let $L := K[\sqrt{\alpha}]$ be the quadratic extension of K obtained by adjoining a root of α . The group $C_{m,\alpha}(K) \setminus \{P\}$ is isomorphic to the subgroup $\{\beta \in L^\times : N_K^L(\beta) = 1\} \subset L^\times$. For instance, when $K = \mathbb{R}$ and $\alpha = -1$, we have that $C_{m,-1}(\mathbb{R}) \setminus \{P\} \cong_{\text{Ab}} S^1$, where $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ is the circle group (draw a picture!). What happens in characteristic two?
- (f) How far can you generalize the above discussion when we don't require K to be perfect or C to be geometrically integral? What works still, and what breaks down?

Exercise 2.6.11. Let K be a field, let $a_1, \dots, a_6 \in K$, let $E \subset \mathbb{P}_K^2$ be the curve defined by [1.9.3](#), let Δ be as in [1.9.2](#) and let c_4 be as in [1.11.14](#).

- (a) The curve E is smooth iff $\Delta \neq 0$. (This was basically shown in [1.9.9](#) the point of this exercise is to ask you to recall this result and fill in all the gaps in the proof left to the reader there.)
- (b) The curve E is nodal iff $\Delta = 0$ and $c_4 \neq 0$. If $\text{ch } K \neq 2$, then E is split nodal iff $\Delta = 0$ and c_4 is a nonzero square. What happens if $\text{ch } K = 2$? (Hint: How does the invariant c_4 transform under admissible changes of coordinates? When $\text{ch } K = 2$, you may have to split into cases depending on whether K is perfect.)
- (c) The curve E is cuspidal iff $\Delta = c_4 = 0$.

This is the geometric meaning of the invariant c_4 corresponding to a Weierstrass equation, at least when the curve it defines is singular.

Exercise 2.6.12. A sequence (b_k) of integers is said to be a *divisibility sequence* if we have that for all $d, n \in \mathbb{Z}_{\geq 1}$ that if $d \mid n$ then $b_d \mid b_n$. It is said to be a *strong divisibility sequence* (also called a *GCD sequence* on the Ross Sets) if for all $m, n \in \mathbb{Z}_{\geq 1}$ we have that $\gcd(b_m, b_n) = b_{\gcd(m, n)}$.

- (a) A GCD sequence is a divisibility sequence.
- (b) For any $b \in \mathbb{Z}_{\geq 1}$, the sequence defined by $b_k := b^k - 1$ is a GCD sequence. The Fibonacci numbers $b_k = F_k$ form a GCD sequence.
- (c) Let $E/\mathbb{Q}$ be an elliptic curve, say $E: y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ with all $a_i \in \mathbb{Z}$. Let $P \in E(\mathbb{Q})$. Define the sequence (b_k) of positive integers by

$$b_k := \begin{cases} 1 & \text{if } kP = \mathcal{O}, \text{ and} \\ e & \text{if } kP = \left(\frac{m}{e^2}, \frac{n}{e^3}\right) \in E(\mathbb{Q}) \setminus \{\mathcal{O}\}, \end{cases}$$

where as in [2.4.14] we have $m, n, e \in \mathbb{Z}$ with $e \geq 1$ and $\gcd(m, e) = \gcd(n, e) = 1$. The sequence (b_k) is a divisibility sequence. Is it a GCD sequence in general?

See also [2.6.20]

Exercise 2.6.13. Let A be a finitely generated abelian group, so that thanks to [1.21.4] A has the form

$$\mathbb{Z}^r \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_s\mathbb{Z}$$

for integers $r, s \in \mathbb{Z}_{\geq 0}$ and $a_1, \dots, a_s \in \mathbb{Z}_{\geq 2}$ such that $a_1 \mid a_2 \mid \cdots \mid a_s$.

- (a) We have that $r = \dim_{\mathbb{F}_p} A/pA$ for all but finitely many primes p . In particular, r is uniquely determined by A .
- (b) The integer $r + s$ is the maximum value of $\dim_{\mathbb{F}_p} A/pA$ as p ranges over all primes. In particular, s is uniquely determined by A .
- (c) Suppose now that $s \geq 1$. The integer a_s is the least positive integer a such that $a \cdot \text{Tors}(A) = 0$. (In other words, a_s is the *annihilator* of $\text{Tors}(A)$.) In particular, a_s is uniquely determined by A .
- (d) (*) More generally, for each $t \in \mathbb{Z}_{\geq 1}$ with $1 \leq t \leq s$, we have that a_t is the least positive integer a such that the subgroup $a \cdot \text{Tors}(A)$ of A can be generated as an abelian group by at most $s - t$ elements. In particular, a_t is uniquely determined by A .

Exercise 2.6.14. Let K be a field and E/K an elliptic curve such that $E(K)$ is finitely generated. This happens, for instance, when K is a finite field, or when $K = \mathbb{Q}$ [1.15.8], or more generally when K is a number field [1.15.8] which we have admittedly not proven. Let $s(E/K)$ denote the number of elementary divisors of the finitely generated abelian group $E(K)$ [1.21.4]. Then we necessarily have $s(E/K) \in \{0, 1, 2\}$, and all three cases occur in general. (Hint: You may use [1.12.27] and [1.12.28] which we also have not fully proven.)

Exercise 2.6.15. For an $n \in \mathbb{Z} \setminus \{0\}$, let $\omega(n)$ denote the number of distinct prime factors of n . Let $a, b \in \mathbb{Z}$ and $(a', b') := (-2a, a^2 - 4b)$ as usual. Suppose that $bb' \neq 0$, so that $E: y^2 = x^3 + ax^2 + bx$ defines an elliptic curve over $\mathbb{Q}$. In this case,

$$\text{rank } E \leq \omega(b) + \omega(b') - 1.$$

This inequality is sharp. How might this generalize to number fields? (Hint: [1.21.2] and obstructions at infinity.)

Exercise 2.6.16. A positive rational number $D \in \mathbb{Q}_{>0}$ is said to be *congruent* iff there is a right-angled triangle with rational sides and area equal to D . Note that this property is visibly invariant under multiplication by rational squares, so we might as well focus on the case of squarefree $D \in \mathbb{Z}_{\geq 1}$. We showed in class [1.22.7] that 1 is not a congruent number. The problem of determining which squarefree positive integers are congruent—also known as the *congruent number problem*—is very old and goes back at least to the 10th century (see the discussion of the history in [13] §1.1). Henceforth, by the term *congruent number*, we mean a *congruent squarefree positive integer*.

- (a) List at least 10 distinct congruent numbers. Which of the integers $D \leq 10$ are congruent? Check that the first few numbers listed in the OEIS sequence A003273 agree with your answer to the previous question.

- (b) There are infinitely many congruent numbers.
 (c) There are infinitely many right-angled triangles with rational sides and area 5.
 (d) (**) The number D is congruent iff the elliptic curve $E_D : y^2 = x^3 - D^2x$ over $\mathbb{Q}$ has positive rank, in which case there are infinitely many right-angled triangles with rational sides and area D .
 (Hint: See 2.3.10, which cites 1 §12].)

For more on the congruent number problem, see 13 §1.1], 25 Chapter 1], and 26.

Exercise 2.6.17. (Adapted from 1 §18, Lemma 1].) Suppose $a, b, c \in \mathbb{Z}_{\geq 2}$ are distinct integers such that $d := abc$ is cubefree. Consider the cubic curves $C_{a,b,c}, D_d, E_d \subset \mathbb{P}_{\mathbb{Q}}^2$ given by

$$C_{a,b,c} = \mathbb{V}(aX^3 + bY^3 + cZ^3), \quad D_d = \mathbb{V}(X^3 + Y^3 - dZ^3), \quad E_d := \mathbb{V}(Y^2Z - X^3 + 432d^2Z^3).$$

If $C_{a,b,c}(\mathbb{Q}) \neq \emptyset$, then $E_d(\mathbb{Q}) \neq \{\mathcal{O}\}$, where $\mathcal{O} = [0 : 1 : 0]$ as usual. (Hint: We met D_d in 2.2.4. We saw in 1.2.B that D_d is isomorphic over $\mathbb{Q}$ to E_d . Now suppose $[X : Y : Z] \in C_{a,b,c}(\mathbb{Q})$. Let ω be a primitive cube root of unity. If $\xi := aX^3 + \omega bY^3 + \omega^2 cZ^3$, then $[\xi : \xi : 3XYZ] \in D_d(\mathbb{Q}[\omega])$. Finally, use 2.1.14.)

Exercise 2.6.18. (For those who know about Hensel's Lemma.)

- (a) The equation $w^2 = 2u^4 - 34v^4$ of 1.22.15 is everywhere locally unobstructed, i.e., has a nontrivial solution (u, v, w) over $\mathbb{Q}_p$ for each prime $p \leq \infty$. In particular, 1.22.15 gives a genuine counterexample to the local-to-global principle.⁵
 (b) In the notation of 2.6.17 let $C = C_{3,4,5} \subset \mathbb{P}_{\mathbb{Q}}^2$; this is the curve of 1.2.7. The curve C is everywhere locally unobstructed, i.e., for each prime $p \leq \infty$, we have $C(\mathbb{Q}_p) \neq \emptyset$.⁶
 (c) (**) (This needs algebraic number theory.) The curve C of (b) has no rational points, i.e., $C(\mathbb{Q}) = \emptyset$. In particular, C is another counterexample to the local-to-global principle. (Hint: Use two-descent as in 2.5.17 to compute the rank of E_{60} and hence find all rational points on it.)

2.6.C Challenge Problems

Exercise 2.6.19. (Adapted from 8 Exercises 6.12-13]) For $k \in [0, 1)$, define the *complete elliptic integrals to the modulus k* to be

$$K(k) := \int_0^1 \frac{du}{\sqrt{(1-u^2)(1-k^2u^2)}} \text{ and } T(k) := \int_0^1 \sqrt{\frac{1-k^2u^2}{1-u^2}} du.$$

- (a) We showed in 1.12.A that the ellipse $C_{a,b} : (x/a)^2 + (y/b)^2 = 1$ with $a \geq b > 0$ has perimeter $P = 4aT(e)$, where $e := \sqrt{1 - (b/a)^2}$ denotes the eccentricity of $C_{a,b}$. The integral $K(k)$ can similarly be expressed in terms of the ellipse $C_{a,b}$; namely, if we let

$$I(a, b) := \int_0^{\pi/2} \frac{dt}{\sqrt{a^2 \cos^2 t + b^2 \sin^2 t}},$$

then $I(a, b) = a^{-1} \cdot K(e)$.

- (b) (*) For all $a, b \in \mathbb{R}$ with $a \geq b > 0$,

$$I(a, b) = I\left(\frac{a+b}{2}, \sqrt{ab}\right).$$

In particular, K satisfies the functional equation

$$K\left(\frac{2\sqrt{k}}{1+k}\right) = (1+k)K(k).$$

(Hint: Use the substitution $x = b \tan t$ to express $I(a, b)$ as the integral of a function from 0 to ∞ . Then use the substitution $x = y + \sqrt{y^2 + ab}$. A different substitution achieving this—due to Gauss—can be found in 28.)

⁵If you get stuck, a solution can be found in 18. Please attempt the problem yourself first!

⁶If you get stuck, a solution can be found in 27. Please attempt the problem yourself first!

- (c) Let S be the arclength of the lemniscate $r^2 = \cos(2\theta)$. Then

$$S = 2\sqrt{2} \cdot K\left(\frac{1}{\sqrt{2}}\right) = 4 \int_0^1 \frac{dx}{\sqrt{1-x^4}}.$$

- (d) For initial values $a, b \in \mathbb{R}$ with $a \geq b > 0$, we recursively define the sequences $\{a_n\}$ and $\{b_n\}$ by $(a_0, b_0) := (a, b)$ and

$$(a_{n+1}, b_{n+1}) := \left(\frac{a_n + b_n}{2}, \sqrt{a_n b_n}\right)$$

for $n \in \mathbb{Z}_{\geq 0}$. Then that

$$a \geq a_1 \geq a_2 \geq \cdots \geq b_2 \geq b_1 \geq b,$$

and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n$. This limit is called the *arithmetic-geometric mean* (AGM) of a and b , and is denoted $\text{agm}(a, b)$. Write a simple program in the language of your choice which takes as input two real numbers a, b as above and returns $\text{agm}(a, b)$, and use this to compute $\text{agm}(a, b)$ for various values of a and b . How rapid do you observe the convergence $a_n, b_n \rightarrow \text{agm}(a, b)$ to be? Make and prove conjectures.

- (e) For $a, b \in \mathbb{R}$ with $a \geq b > 0$, we have that

$$\text{agm}(a, b) \cdot I(a, b) = \frac{\pi}{2}.$$

In particular, for any $k \in [0, 1)$ we have that

$$K(k) = \frac{\pi}{2} \cdot \frac{1}{\text{agm}(1, \sqrt{1-k^2})}.$$

This can be used to compute $K(k)$ very efficiently. Finally, we have

$$\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\pi}{2} \cdot \frac{1}{\text{agm}(\sqrt{2}, 1)}.$$

According to [8], “the observation that these two numbers, calculated independently, agreed to eleven decimal places led Gauss to initiate an extensive study of the arithmetic-geometric mean”. For more details, see [8] Exercises 6.12-13] and [28].

Exercise 2.6.20. (Adapted from [5] Exercise 2.6.11] and [8] Exercises 3.34-3.36].) Let K be a field. A *nondegenerate elliptic divisibility sequence* (EDS) over K is a sequence (b_k) of elements of K determined by four initial parameters b_1, b_2, b_3, b_4 with $b_1 b_2 b_3 \neq 0$ subject to the recursive relations

$$b_{2k+1} = \frac{1}{b_1^3} (b_{k+2} b_k^3 - b_{k-1} b_{k+1}^3), \text{ and}$$

$$b_{2k} = \frac{1}{b_1^2 b_2} b_k (b_{k+2} b_{k-1}^2 - b_{k-2} b_{k+1}^2)$$

for all $k \geq 2$.

- (a) The identity sequence $b_k = k$ is an EDS. The sequence $b_k = F_{2k-1}$ of the odd-indexed Fibonacci numbers is an EDS. More generally, given $b_1, b_2, u, v \in K$, the sequence (b_k) defined by the linear recursive relation

$$b_k = u b_{k-1} + v b_{k-2}$$

for $k \geq 2$ is an EDS. (Hint: Find a subsequence as for the Fibonacci numbers that *is* an EDS.)

- (b) If (b_k) is an EDS, then for each $m \geq 1$ such that $b_m \neq 0$, so is the sequence $(b_{mk}/b_m)_k$. An EDS such that $b_1 = 1$ is said to be *normalized*; given any sequence b we define its *normalization* $\tilde{b}$ to be given by $\tilde{b}_k = b_k/b_1$ for $k \geq 1$. Given a normalized EDS (b_k) , we define its *discriminant* to be

$$\Delta := b_4 b_2^{15} - b_3^3 b_2^{12} + 3b_4^2 b_2^{10} - 20b_4 b_3^3 b_2^7 + 3b_4^3 b_2^5 + 16b_3^6 b_2^4 + 8b_4^2 b_3^2 b_2^2 + b_4^4.$$

We say that a EDS is *singular* if the discriminant of its normalization is zero; else it is said to be *nonsingular*. Which of the sequences from (a) are nonsingular?

- (c) Assume for simplicity that $\text{ch } K \neq 2, 3$. Let $E : y^2 = x^3 + bx + c$ be an elliptic curve over K , and let $P = (x_0, y_0) \in E(K)$. The sequence (b_k) defined by

$$b_k = \begin{cases} f_k(x_0) & k \text{ odd, and} \\ 2y_0 \cdot f_k(x_0), & k \text{ even,} \end{cases}$$

is an EDS, where the polynomials f_k are as in Exercise [2.2.10](#). What is the discriminant of (the normalization of) this sequence b_k ? Is this sequence singular?

- (d) The sequence (b_k) is an EDS iff for each $m > n > r > 0$, we have

$$b_{m+n}b_{m-n}b_r^2 = b_{m+r}b_{m-r}b_n^2 - b_{n+r}b_{n-r}b_m^2.$$

- (e) Now suppose that $K = \text{Frac } R$ for some integral domain R , and let (b_k) be an EDS over K such that $b_1, b_2, b_3, b_4 \in R$ and such that $b_1 \mid b_k$ for $k = 2, 3, 4$ and $b_2 \mid b_4$. Then (b_k) is a divisibility sequence in the sense of [2.6.12](#). If, further, R is a PID and $\gcd(b_3, b_4) = 1$, then (b_k) is a GCD sequence. In particular, these properties hold for the Fibonacci sequence F_k .
- (f) Finally suppose that $K = \mathbb{R}$. Suppose that (b_k) is a nonsingular non-periodic EDS. Then there is a real number $h > 0$ such that

$$\lim_{n \rightarrow \infty} \frac{\log |b_n|}{n^2} = h.$$